Lecture 11

Maxwell's constraints

The Klein–Gordon Legendre map determines the velocity from the momentum. The Maxwell density contains no time derivative of the scalar potential. Its momentum cannot determine that velocity. We retain the undetermined velocity and require the momentum relation to hold throughout evolution. This requirement produces Gauss's law.

Objectives. *We derive Maxwell's primary and secondary constraints and their preservation. We prove that the smeared Gauss constraint generates a spatial gauge transformation and distinguish it from the full time-dependent transformation.*

By the end of this lecture you will be able to:

- Identify the singular velocity Hessian and its primary constraint.
- Derive and propagate Gauss's law with a prescribed conserved source.
- Compute the constraint brackets and the generated gauge variation.

11.1 The failed Legendre inverse

Definition 11.1 (Singular Lagrangian and primary constraint)**.** For velocity coordinates v^a, the *velocity Hessian* of a density is $W_{ab} = \partial^2 L/\partial v^a \partial v^b$. The density is *singular* where this matrix is not invertible. A *primary constraint* is a relation on the canonical variables that holds on the image of the Legendre map.

Use the component Maxwell density derived in Lecture 9:

$$L = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 - \rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A}, \qquad \mathbf{E} = -\mathbf{A}_t - \nabla\phi_{\mathrm{em}}, \qquad \mathbf{B} = \operatorname{curl}\mathbf{A}. \tag{11.1}$$

The prescribed source is held fixed in every variation. The canonical momenta are

$$\pi_\phi = \frac{\partial L}{\partial(\partial_t\phi_{\mathrm{em}})} = 0, \qquad \mathbf{\Pi} = \frac{\partial L}{\partial \mathbf{A}_t} = \mathbf{A}_t + \nabla\phi_{\mathrm{em}} = -\mathbf{E}. \tag{11.2}$$

In the velocity order $(\partial_t\phi_{\mathrm{em}}, A_{x,t}, A_{y,t}, A_{z,t})$, the Hessian is

$$W = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{11.3}$$

Its kernel contains $(1, 0, 0, 0)$ and its rank is three. The relation $\pi_\phi = 0$ is primary. Only the spatial velocity is determined: $\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}$. Changing $\partial_t\phi_{\mathrm{em}}$ leaves all four momenta unchanged. No notation for an inverse matrix can recover this missing velocity. Adding a term $\frac{1}{2}(\partial_t\phi_{\mathrm{em}})^2$ would remove that kernel by changing the density and its equations.

11.2 Extended canonical variables

We work on $\mathbb{R}^3$ with real Schwartz fields. The extended phase space is

$$\mathcal{Z}_{\mathrm{em}} = \{(\phi_{\mathrm{em}}, \mathbf{A}; \pi_\phi, \mathbf{\Pi}) : \text{all eight components lie in } \mathcal{S}(\mathbb{R}^3, \mathbb{R})\}.$$

The scalar momentum is retained as an independent coordinate before imposing its constraint. This space includes the two canonical pairs $(\phi_{\mathrm{em}}, \pi_\phi)$ and $(\mathbf{A}, \mathbf{\Pi})$, the latter with three components. Admissibility means the same as in Lecture 7: a smooth real functional has functional derivatives forming a smooth Schwartz-valued gradient map. Variations pair with those derivatives by the sum of the eight real L^2 pairings.

Definition 11.2 (Extended field bracket)**.** For admissible F and G on $\mathcal{Z}_{\mathrm{em}}$, their bracket is

$$\{F, G\} = \int_{\mathbb{R}^3} \left[F_{\phi_{\mathrm{em}}} G_{\pi_\phi} - F_{\pi_\phi} G_{\phi_{\mathrm{em}}} + F_{\mathbf{A}} \cdot G_{\mathbf{\Pi}} - F_{\mathbf{\Pi}} \cdot G_{\mathbf{A}}\right] \mathrm{d}^3x. \tag{11.4}$$

This is the sum of four brackets of Lecture 7. The same constant canonical matrix and symmetric-Hessian proof establish closure and Jacobi. All bracket statements below concern these admissible functionals or their smeared versions, rather than point evaluations.

Let the source ρ and $\mathbf{J}$ be smooth time-dependent Schwartz functions on an open time interval I, with

$$\rho_t + \operatorname{div} \mathbf{J} = 0. \tag{11.5}$$

We consider compatible initial data in this phase space. They necessarily have zero total charge: integrating $\operatorname{div} \mathbf{\Pi} + \rho = 0$ and using Schwartz decay gives $\int \rho = 0$. Vacuum is included. Nonzero total charge requires a larger spatial class, as in the Coulomb-tail question of Sheet 10.

11.3 The Hamiltonian and preservation of constraints

The Legendre subtraction on the primary surface gives the *canonical Hamiltonian* $H_c : \mathcal{Z}_{\mathrm{em}} \to \mathbb{R}$:

$$H_c = \int_{\mathbb{R}^3} \left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\mathbf{B}|^2 - \mathbf{\Pi} \cdot \nabla\phi_{\mathrm{em}} + \rho\phi_{\mathrm{em}} - \mathbf{J} \cdot \mathbf{A}\right] \mathrm{d}^3x. \tag{11.6}$$

Indeed, substitute $\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}$ in $\mathbf{\Pi} \cdot \mathbf{A}_t - L$. We extend this formula off the primary surface by leaving it independent of π_ϕ.

The boundary term in the scalar-potential term is visible on a cube Q_R:

$$-\int_{Q_R} \mathbf{\Pi} \cdot \nabla\phi_{\mathrm{em}} \, \mathrm{d}^3x = \int_{Q_R} \phi_{\mathrm{em}} \operatorname{div} \mathbf{\Pi} \, \mathrm{d}^3x - \int_{\partial Q_R} \phi_{\mathrm{em}} \mathbf{\Pi} \cdot n \, \mathrm{d}S.$$

Schwartz decay makes the face integral tend to zero. Thus

$$\begin{aligned} H_c &= H_0 + \int \phi_{\mathrm{em}} (\operatorname{div} \mathbf{\Pi} + \rho) \, \mathrm{d}^3x - \int \mathbf{J} \cdot \mathbf{A} \, \mathrm{d}^3x, \\ H_0 &= \frac{1}{2} \int (|\mathbf{\Pi}|^2 + |\mathbf{B}|^2) \, \mathrm{d}^3x. \end{aligned} \tag{11.7}$$

The coefficient of the scalar potential will be the secondary constraint.

Definition 11.3 (Total Hamiltonian and secondary constraint)**.** For an arbitrary prescribed smooth Schwartz multiplier $\lambda(t,x)$, the *total Hamiltonian* is

$$H_T(t) = H_c(t) + \int_{\mathbb{R}^3} \lambda(t,x)\pi_\phi(x)\,\mathrm{d}^3x. \tag{11.8}$$

Evolution obeys $\dot F = \partial_t F + \{F, H_T\}$. A *secondary constraint* is a new relation required to preserve a primary constraint during this evolution.

The extra term assigns $\partial_t \phi_{\mathrm{em}} = \lambda$ without imposing a new relation between that velocity and its zero momentum. The Hamiltonians above have smooth Schwartz gradients for the stated fields, sources and multiplier.

Proposition 11.4 (Gauss constraint and its generators)**.** *The primary constraint $\pi_\phi = 0$ is preserved exactly when the* Gauss constraint *holds:*

$$\mathcal{D} := \operatorname{div}\mathbf{\Pi} + \rho = 0. \tag{11.9}$$

This secondary constraint propagates under (11.5)*. For fixed real $\zeta, \chi \in \mathcal{S}$, define the smeared constraints*

$$P[\zeta] = \int \zeta\pi_\phi\,\mathrm{d}^3x, \qquad C[\chi] = -\int \chi(\operatorname{div}\mathbf{\Pi} + \rho)\,\mathrm{d}^3x. \tag{11.10}$$

Their mutual brackets vanish identically. The generator $C[\chi]$ shifts $\mathbf{A}$ by $\nabla\chi$ and leaves $\mathbf{\Pi}$ unchanged.

Proof. We compute the Hamilton equations before restricting to the constraints. Varying the magnetic term gives

$$\delta\left(\frac{1}{2}\int_{Q_R} |\mathbf{B}|^2\,\mathrm{d}^3x\right) = \int_{Q_R} (\operatorname{curl}\mathbf{B})\cdot\delta\mathbf{A}\,\mathrm{d}^3x + \int_{\partial Q_R} n\cdot(\delta\mathbf{A}\times\mathbf{B})\,\mathrm{d}S.$$

This follows by integrating the coordinate curl derivatives by parts. The boundary integral vanishes as $R\to\infty$ by Schwartz decay. Differentiating the other terms of H_T gives

$$\partial_t\phi_{\mathrm{em}} = \lambda, \qquad (\pi_\phi)_t = -\mathcal{D}, \qquad \mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\mathbf{B} + \mathbf{J}. \tag{11.11}$$

These are equations for smooth Schwartz paths; equivalently, each follows by testing the bracket against arbitrary Schwartz smearings. The second equation proves the primary-preservation assertion. The last equation and commutation of spatial derivatives give

$$\mathcal{D}_t = \operatorname{div}\mathbf{\Pi}_t + \rho_t = \operatorname{div}\mathbf{J} + \rho_t = 0, \tag{11.12}$$

since the divergence of a curl is zero. There is no further constraint. The equations for $\mathbf{A}$ and $\mathbf{\Pi}$, with $\mathbf{E} = -\mathbf{\Pi}$, are Ampère's law and the potential relation of Lecture 9. The secondary constraint is Gauss's law.

Next, integrate the divergence in $C[\chi]$ by parts. Its only nonzero phase derivative is $C_{\mathbf{\Pi}} = \nabla\chi$. The only nonzero derivative of $P[\zeta]$ is $P_{\pi_\phi} = \zeta$. They are constant Schwartz gradient maps. Formula (11.4) gives

$$\{P[\zeta], P[\vartheta]\} = 0, \qquad \{P[\zeta], C[\chi]\} = 0, \qquad \{C[\chi], C[\vartheta]\} = 0. \tag{11.13}$$

For the generator assertion, its Hamiltonian vector has spatial components $\delta\mathbf{A} = C_{\mathbf{\Pi}} = \nabla\chi$ and $\delta\mathbf{\Pi} = -C_{\mathbf{A}} = 0$. Its scalar components are both zero. ∎

The source is explicitly time dependent in $C[\chi]$. Its preservation must include that dependence. For fixed smearings the same calculation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}P[\zeta] = \{P[\zeta], H_T\} = C[\zeta], \qquad \frac{\mathrm{d}}{\mathrm{d}t}C[\chi] = \partial_t C[\chi] + \{C[\chi], H_T\} = -\int \chi(\rho_t + \operatorname{div}\mathbf{J})\,\mathrm{d}^3x = 0.$$

Ignoring $\partial_t C$ would lose the ρ_t term.

Definition 11.5 (Weak equality and constraint classes)**.** The *constraint surface* Γ_t is the common zero set of all $P[\zeta]$ and $C[\chi]$ at time t. The notation $F \approx G$ means equality on Γ_t; this is *weak equality*. A *strong equality* holds on the entire extended phase space. A constraint is *first class* if its bracket with every defining constraint vanishes weakly. For a finite list of constraints, the list is *second class* if its bracket matrix is invertible on the constraint surface.

Since π_ϕ and $\mathcal{D}$ are Schwartz, their squared-norm tests show that Γ_t is exactly the set $\pi_\phi = \mathcal{D} = 0$. All Maxwell constraints above are first class, because (11.13) holds strongly. In contrast, $\{P[\zeta], H_T\} = C[\zeta]$ vanishes only weakly in general. No inverse bracket matrix is asserted for field constraints here.

11.4 The smeared gauge transformation

For $\chi \in \mathcal{S}$, consider the admissible functional

$$G[\chi] = \int_{\mathbb{R}^3} \mathbf{\Pi} \cdot \nabla\chi \,\mathrm{d}^3x. \tag{11.14}$$

On Q_R, its relation to the Gauss constraint includes the boundary term:

$$C_{Q_R}[\chi] = G_{Q_R}[\chi] - \int_{Q_R} \chi\rho \,\mathrm{d}^3x - \int_{\partial Q_R} \chi\mathbf{\Pi}\cdot n \,\mathrm{d}S.$$

The last integral vanishes for our Schwartz data. Hence $G[\chi] = C[\chi] + \int \chi\rho \,\mathrm{d}^3x$. The source term is constant as a function of the canonical variables. These functionals have the same Hamiltonian vector field, although $G[\chi]$ is not generally zero on Γ_t.

For a real vector test function $f \in \mathcal{S}^3$, write $\mathbf{A}(f) = \int f \cdot \mathbf{A}\,\mathrm{d}^3x$ and similarly for $\mathbf{\Pi}(f)$. Their brackets compute the transformation entirely inside the admissible class:

$$\{\mathbf{A}(f), G[\chi]\} = \int f\cdot\nabla\chi \,\mathrm{d}^3x, \qquad \{\mathbf{\Pi}(f), G[\chi]\} = 0, \qquad \{G[\chi], G[\vartheta]\} = 0. \tag{11.15}$$

The flow is exactly $(\mathbf{A}, \mathbf{\Pi}) \mapsto (\mathbf{A} + \varepsilon\nabla\chi, \mathbf{\Pi})$. The curl of a gradient is zero, so both electromagnetic fields are unchanged when the scalar potential is also fixed and χ is time independent.

For $\chi \in C^\infty(I;\mathcal{S})$, the secondary constraint alone still leaves ϕ_{em} fixed. The complete potential transformation of Lecture 9 also requires $\delta\phi_{\mathrm{em}} = -\chi_t$. On the extended phase space its generator is

$$C[\chi] - P[\chi_t].$$

The primary term gives $\delta\phi_{\mathrm{em}} = -\chi_t$ by the canonical bracket. The spatial shift remains $\delta\mathbf{A} = \nabla\chi$ and both momenta stay fixed.

To map Hamiltonian solutions, also transform the multiplier as $\lambda' = \lambda - \partial_t^2\chi$. Indeed, $\partial_t\phi'_{\mathrm{em}} = \partial_t\phi_{\mathrm{em}} - \partial_t^2\chi = \lambda'$. The spatial equation is preserved since $\partial_t\mathbf{A}' = \partial_t\mathbf{A} + \nabla\partial_t\chi = \mathbf{\Pi} - \nabla\phi'_{\mathrm{em}}$; the momentum equations and constraints are unchanged. Thus a spatial generator, a fixed-time phase transformation and a transformation of Hamiltonian solutions have distinct scopes.

Summary. Maxwell's velocity Hessian has rank three, with primary constraint $\pi_\phi = 0$ and spatial momentum $\mathbf{\Pi} = -\mathbf{E}$. The total Hamiltonian retains an arbitrary scalar-potential velocity. Primary preservation gives $\operatorname{div}\mathbf{\Pi} + \rho = 0$, whose propagation requires source conservation. The smeared constraints commute strongly and are first class. The Gauss generator shifts $\mathbf{A}$ by $\nabla\chi$; its difference from $G[\chi]$ is the prescribed source term. Lecture 12 removes this spatial gauge freedom in vacuum.

Self-check. Two potentials have the same spatial velocities and spatial derivatives, but their scalar-potential velocities differ. Can their canonical momenta coincide? Which term in H_T retains this freedom?

Answer. Yes. The density contains no $\partial_t\phi_{\mathrm{em}}$, so all four momenta coincide. The term $\int \lambda\pi_\phi$ assigns the arbitrary velocity $\partial_t\phi_{\mathrm{em}} = \lambda$.

Self-check. Is $\{P[\zeta], H_T\} = 0$ a strong identity? What must hold on the constraint surface?

Answer. The bracket equals $C[\zeta]$, which is not identically zero on phase space. It vanishes on Γ_t. This is weak preservation of the primary constraint.

Self-check. At a fixed time, suppose $\int \chi\rho\,\mathrm{d}^3x = 3$. Compare $C[\chi]$ and $G[\chi]$ on Γ_t. Do they generate different spatial variations?

Answer. The constraint has value zero and $G[\chi] = 3$. Their phase derivatives coincide because the source is prescribed, so both generate $\delta\mathbf{A} = \nabla\chi$ and $\delta\mathbf{\Pi} = 0$.

Self-check. For time-dependent χ, which term supplements $C[\chi]$ to generate $\delta\phi_{\mathrm{em}} = -\chi_t$? Does it alter $\mathbf{\Pi}$?

Answer. The term is $-P[\chi_t] = -\int \chi_t\pi_\phi\,\mathrm{d}^3x$. It shifts the scalar potential and leaves $\mathbf{\Pi}$ unchanged.